Practical Value of an Automated Liver Tumor Segmentation Service

The automated service under development in this project is designed for use in a clinical setting to assist physicians - primarily radiologists - in processing CT scans, thereby enabling them to interpret scans for a larger number of patients, which in turn reduces diagnostic waiting time.

For example, the service can be applied in the following use cases:

1) **Preoperative Resection Planning.** In preparation for liver tumor resection *Future Liver Remnant* (the volume of functional liver tissue remaining post-surgery) must be calculated. The automated segmentation service provides necessary organ and lesion volume calculations, allowing physicians to determine quickly whether resection is a viable option.
2) **Minimally Invasive Interventions Guidance.** In certain procedures such as thermal ablation for colorectal liver metastases, physicians can use the service to accurately determine ablation zones, thereby minimizing damage to healthy tissue.
3) **Monitoring.** For patients undergoing systemic therapy (e.g., chemotherapy) the automated segmentation service provides an efficient way for assessing treatment response.

Several groups can directly benefit from this service:

1) **Radiologists**. The service helps radiologists significantly reduce the amount of time allocated for one CT scan without compromising diagnostic accuracy. Furthermore, the service may help identify suspicious regions that would require additional manual review by a radiologist.
2) **Surgeons**. Surgical teams can use the service to make better-informed, patient-specific operative decisions.
3) **Patients**. Patients experience reduced waiting time between undergoing a CT scan and receiving diagnostic results, thereby accelerating treatment start.

While deploying this service within a hospital or medical center infrastructure is feasible and has substantial potential to improve healthcare quality and efficiency, several challenges (of moderate to high severity) still exist:

1) **Data availability and representativeness.** Most publicly available datasets contain limited sample sizes, constraining the ability to train sophisticated models. Limited training diversity may lead to poor generalization encountering rare anatomical variants or atypical tumor patterns.
   Implementing a *"human-in-the-loop"* approach - wherein a radiologist reviews and verifies model segmentation - can address this issue. However, this option has a different flaw: automation bias.
   Alternatively, synthetic data could be introduced to enrich training dataset.
2) **Computational constraints.** Processing 3D images requires substantial GPU memory. Acquiring a certain number of GPUs to enable local inference may not be cost-effective for healthcare providers. In contrast, choosing cloud inference raises data security concerns.

3) **Legal liability.** Regardless of model accuracy, errors cannot be fully eliminated. The debate centers on accountability when misdiagnoses occur. A sensible approach is to assign liability based on the context of a situation. For example, if the model segments lesion boundaries for surgery, but the surgeon observes that the tissue marked as “tumor” is healthy, the surgeon has the duty to avoid resecting healthy parenchyma.

In order to evaluate model performance it is possible to use standard segmentation metrics (such as Dice, IoU, Hausdorff Distance, Recall, etc.) on test datasets. However, good metrics on test dataset do not automatically guarantee real-world clinical value. Consequently, additional validation by physicians is essential. A sound approach may involve evaluating CT scans by both the model and independently by radiologists over a period of time. This experiment will provide evidence regarding the practical value of the service in a particular clinical environment.